What Time Use Surveys Can (And Cannot) Tell Us About Labour Supply

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This version: November 2019

Abstract

Nonclassical measurement errors in conventional microeconomic surveys result in biased estimates regarding weekly labour supply. The American Time Use Survey (ATUS) accurately measures hours worked on a single day. We show that weekly labour supply parameters can be consistently and efficiently estimated using the ATUS, but it is impossible to recover weekly hours or its distribution. We propose *impute estimator* and carefully examine its properties. It is a simple modification of the 2SLS estimator, which imputes both dependent and independent variables using daily subsamples. We apply it to the ATUS and find substantially different elasticity estimates from the CPS, especially for married women.

Keywords: labour supply, time specificity, measurement error, survey methods, instrumental variable, asymptotic efficiency, standard error
JEL codes: J22, C21

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1 Introduction

Empirical studies of labour supply depend greatly upon data on how much time people spend working. Unfortunately, there is abundant evidence showing that weekly hours worked are poorly measured in frequently used survey data sets such as the Current Population Survey (CPS) and the Panel Study of Income Dynamics (PSID), and that the measurement error is nonclassical (for example, Bound, Brown, Duncan and Rodgers, 1989; Bound, Brown and Mathiowetz, 2001). This nonclassical measurement error in weekly hours worked is found to significantly bias the estimation of labour supply parameters (Barrett and Hamermesh, 2019).

On the other hand, since economists have been interested in how people allocate their time on market work and nonmarket activities (Mincer, 1962; Becker, 1965), many countries have historical or ongoing time use surveys.[1] Time use surveys typically ask the respondents to record all their activities during a prescribed period (one or a few days) in the format of a detailed diary, and hence provide relatively accurate measure of how individuals allocate their time (for example, Knabe, Rätzel, Schöb and Weimann, 2010; Luengo-Prado and Sevilla, 2012; Aguiar, Bils, Charles and Hurst, 2017).[2], Since the measurement errors in time use surveys are negligible compared to traditional surveys, it is natural to ask whether and how time use surveys can be utilized to quantify and/or rectify the empirical consequences of error-ridden weekly hours worked in traditional surveys like the CPS.

The major difficulty in deploying time use surveys to rectify error-ridden weekly hours worked in the CPS is that time use surveys typically provide information about labour supply for only a few days, but not all days of a week. For example, the American Time Use Survey (ATUS) records one single day for each respondent, and the UK time use survey records two days. To the best of our knowledge, the only exception is the Dutch Time Use Survey

[1] These countries include Australia, Canada, China, Japan, New Zealand, Pakistan, Russia, the USA and most European countries.

[2] For a review of time use surveys used for studies in other subfields of economics, see Aguiar, Hurst and Karabarbounis (2012).

(DTUS),[3] in which respondents record their activities for seven consecutive days. If we are interested in weekly labour supply, then ideally we need to observe typical weekly hours worked. The single day observed in the ATUS, albeit randomly sampled, creates a missing data problem or "time specificity" problem, as Barrett and Hamermesh (2019) put it.

The goal of this paper is to study the consequences and to provide solution of the time specificity problem from the viewpoint of labour supply, and we find the best way to leverage the daily hours in time use surveys to estimate labour supply. First, we employ the potential outcome framework to show that the time specificity problem results in several impossibility results, whose empirical consequences have not been fully appreciated. Second, the main methodological contribution of this paper is to prove that despite the impossibility results, the daily hours worked in the ATUS can be utilized to consistently and efficiently estimate weekly labour supply equation under the same condition as if the *true* weekly hours worked were observed.[4] We propose several easy-to-implement consistent estimators, and recommend what we name *impute estimator* based on its superior asymptotic efficiency and finite sample stability. The impute estimator is a simple modification of the standard two stage least squares (2SLS) estimator, where the same instrumental variables (IVs) are used to impute the (unobserved) dependent variable as well as the (endogenous) independent variables. Our impute estimator essentially matches similar individuals based on the IVs, and uses observed hours of matched people as imputed hours for those who were not surveyed by the ATUS on a particular day. Our analysis using the DTUS as the benchmark strongly supports the superiority of the impute estimator as well, and it reveals immense biases in labour supply elasticity estimates as a result of measurement errors in traditional survey type recalled weekly hours. Third, using a sample of American population who participated in both ATUS and CPS,

[3] In Dutch, it is called Het Tijdsbestedingsonderzoek. In this paper we call it the DTUS for the consistency with the ATUS.

[4] If the true weekly hours worked were observed and the regressors (e.g. wage) are endogenous, then the usual 2SLS estimator only requires valid and relevant IVs to be available. As becomes clear below, the argument generalizes trivially to other time use surveys for more than one day.

we uncover multiple interesting empirical findings by applying our impute estimator to the ATUS and comparing the estimates with those given by the CPS. We find that the ATUS yields smaller own wage elasticities than the CPS across the board, but the gaps vary among gender and marital groups. Moreover, the ATUS indicates smaller spouse (cross) earning elasticity than the CPS for married women, but larger for married men. Furthermore, the ATUS exhibits weaker elasticity with respect to number of older kids than the CPS for married women, even though the two surveys result in almost the same elasticity estimates with respect to number of younger kids.

To understand the time specificity problem, we are the first to employ the potential outcome framework to time use surveys. The hours worked on each day of a week can be thought of as one potential outcome. Weekly hours worked are the sum of the seven potential outcomes. For each individual, the ATUS provides information for only one single day, but not the other six. Just like individual treatment effects (differences between treatment outcomes and control outcomes for the same individuals) cannot be recovered even from purely random experiment data, individual weekly hours worked cannot be retrieved from the ATUS data without ad hoc assumptions. By similar argument, it is easy to see that even the distribution function of weekly hours worked is not identified unless the hours worked on seven days are independent, which is unlikely.[5] Hence, accurate daily hours worked in the ATUS data do not directly quantify or rectify the measurement errors in the weekly hours worked in common surveys like the CPS. In addition, as we will elaborate in Section 3.2.2, the time specificity problem also causes some technical difficulties in computing the standard error of the impute estimator. These impossibility results highlight the limitation of time use surveys.

Our investigation into labour supply parameters, on the other hand, portrays a much brighter picture. Labour supply parameters are often estimated using certain regression function, which is in essence a conditional mean func-

[5]If the daily hours are independent, then the marginal distributions of daily hours are identified from the ATUS data, and the joint distribution of seven daily hours is the product of the marginal distributions, so the distribution of weekly hours is determined by convolution. If the daily hours are not independent, the joint distribution of seven daily hours is not pinned down by their marginal distributions.

tion. Just like that average treatment effects can be identified using conditional mean functions, many important parameters such as labour supply elasticities can be uncovered using time use survey data. In the context of labour supply elasticities, the potential outcome framework serves as a powerful scheme that enables us to propose a number of consistent estimators based on time use surveys, to rigorously examine their asymptotic and finite sample properties (i.e. consistency, asymptotic normality, relative asymptotic efficiency and numerical stability), and to apply the best (impute) estimator to the ATUS so as to empirically rectify the error-ridden estimates based on the CPS. None of these has ever been done in the literature.

We are not the first to recognize the time specificity problem when using the ATUS to estimate weekly labour supply. For example, our impute estimator resembles the "synthetic time diary" approach in Aguiar, Bils, Charles and Hurst (2017) at first glance, but we emphasize that the consistency of all the estimators demands the matching (or equivalently imputation) to be based only on exogenous IVs of the main regression. This seems to challenge the wisdom that one should use all relevant (and likely endogenous) variables in order to make better imputation. Barrett and Hamermesh (2019)'s approach corresponds to the pool estimator in our paper. Nevertheless, we are the first to conduct thorough inspection of the time specificity problem, rigorously prove impute estimator's superior efficiency and robustness, work out standard errors, and uncover new empirical labour supply elasticity estimates.

The time specificity problem can be thought of as a missing data problem. In fact, the hours worked on the non-survey days are missing completely at random and follow the "file-matching" pattern (Little and Rubin, 2012). However, our problem is harder than many file-matching problem since in the ATUS we cannot observe two daily hours for any respondent, and the correlations among daily hours are necessary to obtain the standard errors. Moreover, we take advantage of the special structure of our question at hand and propose an efficient and robust estimator.

Empirical studies have found nonclassical measurement errors in many other dependent variables (Duncan and Hill, 1985; Bound, Brown and Mathiowetz, 2001) as well as labour supply. But in theoretical econometrics lit-

erature, nonclassical measurement errors in dependent variables have drawn far less attention than independent variables (for example, Hu and Schennach, 2008; Chen, Hong and Tamer, 2005; Hu and Sasaki, 2015). A notable exception is Abrevaya and Hausman (1999). While our impute estimator naturally accommodates endogenous independent variables (e.g. wage), it is unclear, however, whether and how Abrevaya and Hausman (1999)'s estimator could be generalized to this case.

As alluded before, the DTUS serves as a valuable benchmark since it contains accurate time use diary hours for seven consecutive days. We randomly draw one single day for each individual to imitate the ATUS. This artificial data set permits direct comparison between our proposed estimators and the usual 2SLS estimator, which is infeasible for the ATUS data. Using this unique creative approach, we manage to unambiguously illustrate the superiority of our impute estimator.

Contemplative readers may wonder: what is the significance of weekly labour supply? Why not estimate monthly, quarterly, or yearly labour supply? The most obvious reason is that the CPS records weekly hours,[6] and we need to aggregate the daily information in the ATUS to weekly in order to compare with the CPS. But more importantly, once we bridge the gap between daily hours and weekly hours, then going from weekly hours to longer time frame follows the exact same logic.[7]

For activities recorded in the ATUS other than working (Aguiar and Hurst, 2007; Guryan, Hurst and Kearney, 2008; Aguiar, Bils, Charles and Hurst, 2017), time specificity problem remains. Time specificity problem also presents itself outside time use surveys, such as recalled food expenditure data (Ahmed, Brzozowski and Crossley, 2006; Sousa, 2014; Brzozowski, Crossley and Winter, 2017) versus the diary system used in the Expenditure and Food Survey (EFS)

[6]The CPS asks the respondents how many hours he/she *usually* works per week, and how many hours he/she *actually* worked the week before, both for their main jobs and other jobs. In our empirical studies using CPS data, we used the number of hours per week that the respondent usually work.

[7]Although Bhuller, Brinch and Königs (2017) find that in dynamic models, which are outside of the scope of this paper, estimation results vary strongly with the time unit of analysis.

in the UK.[8]

The rest of the paper is organized as follows. Section 2 gives more information about time use surveys and traditional surveys. In Section 3, we first give two impossibility results regarding the true weekly hours. Then we focus on the estimation of labour supply parameters. We set up the model and propose several intuitive estimators. We recommend the impute estimator based on our analysis of the asymptotic properties of all the estimators. Section 4 uses the DTUS as the benchmark, and illustrates the superior finite sample properties of the impute estimator. Section 5 applies our impute estimator to the ATUS and compares it with labour supply elasticity estimates produced by the CPS for the same respondents. Our empirical analysis in this section quantifies the biases in various labour supply elasticity estimates due to the measurement errors in the CPS recalled hours for each gender and marital group. Section 6 states a few of our comments on the design of time use surveys. Section 7 concludes the paper.

The Supplementary Appendices collect the proofs, the methods of computing standard errors, and various robustness checks of our empirical studies.

2 Time Use Surveys

The ATUS randomly draws a subsample of the respondents who just completed their participation in the CPS within the past two to five months.[9] On a randomly chosen day (interview day), the respondents are asked to fill up a diary detailing all their activities minute-by-minute on the previous day (diary day). Adding all the time spent on working by each respondent yields his/her ATUS hours worked for the diary day. Since the respondents of the ATUS had already participated in the CPS, all the data collected by the ATUS and the CPS about them are available for analysis, including demographics and income.[10]

[8] The EFS became known as the Living Cost and Food Survey from January 2008.

[9] For the workers who satisfy the criterion for our empirical analysis in Section 5, the number of those who participate in the ATUS account for roughly one fiftieth of all the respondents in the CPS.

[10] For a more detailed description of the ATUS, see Hamermesh, Frazis and Stewart (2005).

The ATUS has some distinct features that set it apart from commonly used surveys like the CPS. First, the respondents of the ATUS recall their activities for only one day (diary day), as opposed to weeks or months. The diary day is completely randomly chosen, with weekends having higher probabilities than weekdays.[11] Second, the ATUS imposes a 24-hour limit on the time allocated to all the recalled activities. These two features are likely to make the ATUS hours a much more accurate measure of the hours worked on a single day. Throughout this paper, we assume that the observed daily hours worked in the ATUS are the true hours worked for the diary day, without any measurement error. We acknowledge that this assumption is almost certainly wrong, and that the incidence and the size of the measurement error in the ATUS daily hours should be carefully examined for any serious empirical research. But it is adequate and convenient for our purpose in this paper.

On the contrary, the CPS records weekly hours, by asking either how many hours the respondents *usually* work per week or how many hours they *actually* worked in the previous week. While probably less accurate than the ATUS hours, the CPS hours concern a longer time period.

In order to quantify and rectify the consequences of error-ridden hours in the CPS using the more accurate ATUS hours, we have to understand and tackle this time specificity of the two data sources. As mentioned before, this time specificity is the crux of this paper. To focus on the consequences of time specificity, we will only include the individuals who participated in both the CPS and the ATUS into our sample for empirical analysis, so that no differences in estimates or efficiency may result from the differences in samples.[12]

Such time specificity of hours between time use surveys and commonly used surveys is not unique to the US, presumably because of high costs of conducting time use surveys. In fact, to the best of our knowledge, the only

[11] The ATUS samples Mondays to Fridays with probability of 0.10 each, and it samples Saturdays and Sundays with probability of 0.25 each. Weekends are oversampled since they are more informative about people's activities other than work.

[12] In Supplementary Appendix B, we compare the ATUS sample with a much bigger CPS sample and do not find significant difference between the distributions of key variables from the two samples.

country that has ongoing time use survey that records activities for an entire week is the Netherlands.[13] The DTUS has been carried out since 1975 and has been published every five years. In the week long diary, the participants record their main activity every ten minutes and a secondary activity that might take place at the same time. The survey randomly draw more than two thousand participants from the Dutch population aged 12 and over since 2006. For the same respondents, the DTUS also contains CPS-type recalled weekly hours and some demographics including age, gender, education and number of children. So the DTUS serves as a particularly precious benchmark against which we can evaluate different estimators. We are going to base our simulation studies on the DTUS. Unfortunately, the DTUS does not contain detailed information on income, which renders it unsuitable for our empirical analysis involving wage or earnings.[14] But the DTUS contains demographic information which allows us to draw some empirical findings about labour supply along that line.[15]

3 Good News and Bad News about Labour Supply

This section has good news and bad news. We start with the bad news—that is, what time use surveys cannot tell us about labour supply. By a very simple and straightforward potential outcome argument, we show that neither the weekly hours worked nor its distribution can be identified using the ATUS type time use survey data. Then we proceed to the good news—that is, time use surveys can provide consistent and relatively efficient estimates of labour supply elasticities under mild conditions.

[13]The Economic and Social Research Council (ESRC) of the UK funded time use survey in 1973, 1974, 1983 and 1984 that covers 7 days of a week, but the more recent time use survey in the UK covers only 2 days.

[14]The income variable in the DTUS is only the annual income quartiles.

[15]For a more detailed description of the DTUS, see Fisher, Gershuny, Flood, Roman and Hofferth (2018).

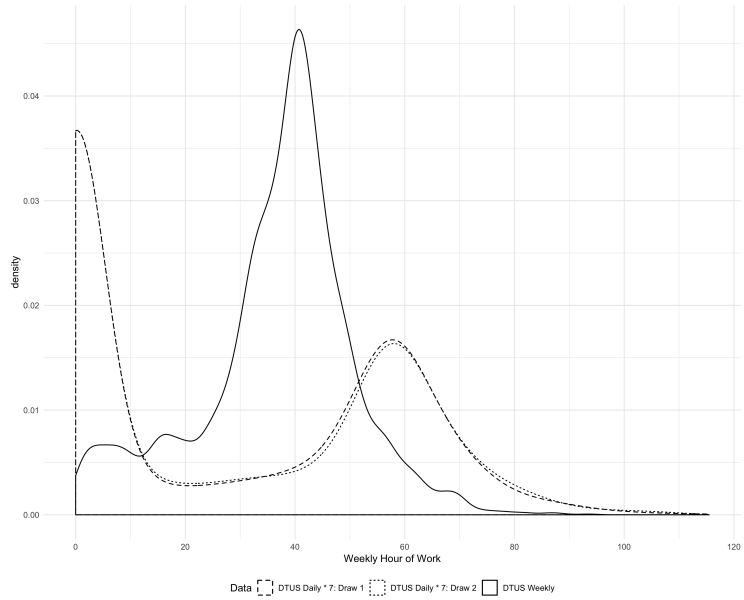

Figure 1: DTUS Weekly Hours vs. Randomly Drawn Daily Hours $\times 7$

3.1 Bad News: Potential Hours and Impossibility Results

Let's start with a simple question: how do we recover the distribution of weekly hours worked from the ATUS daily hours data? Since the ATUS diary day is randomly drawn, one may think of the ATUS daily hours as a representative sample of the weekly hours and, therefore, the distribution of weekly hours may be recovered from the distribution of the ATUS daily hours with adjustment for diary day sampling weights.

A small experiment using the DTUS data illustrates that this is a bad idea. In Figure 1, the solid line shows the kernel density of the DTUS weekly hours worked, which is directly observable in the DTUS for each individual. To mimic the ATUS, we randomly choose one day from the DTUS as the diary day for each individual, and plot the kernel density of the hours worked on the diary day multiplied by 7. The dashed and the dotted lines in Figure 1 show the kernel densities for two such random experiments. They differ from the DTUS weekly hours significantly.[16]

[16] In Supplementary Appendix B, we take the common five-day work schedule into account, and the results are similar.

It turns out that it is just impossible to identify the distribution of weekly hours from daily hours without ad hoc assumptions. Now we introduce some notation to facilitate the discussion. Let the individual respondent be indexed by $i = 1, \ldots, n$. Let H_i^w denote the true weekly hours worked by individual i. The recalled weekly hours worked H_i^{CPS} in the CPS is an error-ridden measure of H_i^w,

$$H_i^{CPS} = H_i^w + e_i. \tag{3.1}$$

That the measurement error is nonclassical implies that e_i could be correlated with H_i^w. Let $t \in \{1, \ldots, 7\}$ denote the days of a week,[17] and let H_{it} denote the true daily hours worked by individual i on day t. Naturally, the weekly hours worked equal to the sum of daily hours worked over the week,

$$H_i^w = \sum_{t=1}^{7} H_{it}. \tag{3.2}$$

Let t_i be the dairy day of individual i in the ATUS, then the daily hours worked in ATUS, denoted as H_i^{ATUS}, is just H_{it_i}. To facilitate our analysis, it helps to write the ATUS daily hours in an alternative way. Let $d_{it} \equiv \mathbb{I}[t_i = t]$ be seven diary day dummy variables for each individual i.[18] Then

$$H_i^{ATUS} = H_{it_i} = \sum_{t=1}^{7} d_{it} H_{it}. \tag{3.3}$$

Since for any individual interviewed in the ATUS, one and only one of the seven diary dummies is one, we only have an accurate measure of his/her hours worked for a single day of the week, but not for the other six days.[19]

Now it helps to recall the conventional wisdom in the program evaluation literature that even in purely random experiments, neither individual treat-

[17] $t = 1$ indicates Sunday, $t = 2$ indicates Monday, and so on.

[18] The symbol $\equiv$ indicates that the quantity on the left side is defined as the expression on the right side.

[19] Throughout the paper, we assume that the hours worked in time use surveys are the true hours worked for the prescribed period. This is merely for the simplicity of exposition, because the main results still hold if the measurement error of hours worked in time use surveys is classical. Estimators will have larger standard errors, but classical measurement errors in dependent variables are known to be much less problematic.

ment effect nor its distribution in the population can be identified without ad hoc assumptions on the joint distribution of (Y_{i1}, Y_{i0}).[20] Following the program evaluation literature, we call H_{it} "potential hours" of diary day t ($t = 1, \ldots, 7$).

Under the potential outcome framework, the following impossibility results naturally follow. First, without ad hoc assumptions, it is impossible to recover individual weekly hours worked H_i^w from what is available in the ATUS. This impossibility result implies some important limitations of $\hat{H}_i^w$, the *imputed* weekly hours we will introduce in the next subsection, and we will discuss them in Remark 1. Second, the ATUS only contains information regarding the marginal distributions of daily hour worked on a single day, but provides no information about the dependence among $(H_{i1}, \ldots, H_{i7})'$. The latter is required to find out the distribution of weekly hours H_i^w. In consequence, the distribution of the weekly hours worked H_i^w (as well as its variance) cannot be recovered using the ATUS daily hours data. Third, computing the standard errors of ATUS based estimators (e.g., Barrett and Hamermesh, 2019) needs extra effort. Because the variance of weekly hours is not known, the conventional standard errors, which require the variance of weekly hours, are not computable.[21] Without the potential outcome framework, this was not obvious, but the reason will become clearer after we give the standard error formulas for various estimators in Theorem 2, Theorem 4 and Remark 8.

3.2 Good News: Labour Supply Parameters

Despite the impossibility results in Section 3.1, daily hours worked in the ATUS nevertheless can produce consistent and relatively efficient parameter estimates in the weekly labour supply regression equation. In particular, such parameters include labour supply elasticities, the application we will focus on in the rest of this paper.

The reason why these parameters in the weekly labour supply regression

[20] Let Y_{i1}, Y_{i0} and d_i denote the outcome if treated, the outcome if not treated and the treatment indicator for individual i, respectively, then the observed outcome is $Y = d_i Y_{i1} + (1 - d_i) Y_{i0}$. It is well known that the individual treatment effect, defined as $Y_{i1} - Y_{i0}$, cannot be identified.

[21] This is consistent with the studies on file-matching patterns of missing data (e.g. Little and Rubin, 2012).

equation can be identified and estimated is again better understood under the potential outcome framework. The ATUS closely resembles purely random experiments, since the diary day is completely randomly chosen for each respondent. In random experiments, $\mathrm{E}(Y_{i1} - Y_{i0})$, the average treatment effect (ATE) and $\mathrm{E}(Y_{i1} - Y_{i0}|X_i = x)$, the conditional average treatment effect (CATE) can be identified and estimated using the data that records either Y_{i1} or Y_{i0} (but not both) for each individual. Since regression equations are essentially conditional mean models, both $\mathrm{E}(H_i^w)$ and $\mathrm{E}(H_i^w|X_i = x)$, the counterparts of the ATE and the CATE in our scenario, can be identified and estimated. In fact, the labour supply elasticity estimator we recommend later in this section resembles a similarity to the matching regression estimator of the average treatment effect in that it uses the actual ATUS daily hours worked by other individuals with similar exogenous characteristics to impute the six missing daily hours worked for each individual in the ATUS.

Two unique features, however, differentiate the labour supply elasticity estimation problem from the usual treatment effect estimation. First, the standard error of our recommended elasticity estimator requires the correlation among the daily potential hours, but the ATUS contains no information regarding it. In order to compute the standard error, we will propose two methods, and the second of which imposes a stronger assumption than necessary for the consistency of the estimator. This stronger assumption essentially requires that the daily potential hours models can be estimated separately, which is automatically assumed in the treatment effect literature (i.e. Y_{i1} and Y_{i0} can be modeled using two separate regression models). Second, elasticity hinges on not only the mean, but also the partial derivative of the conditional mean function $\partial\,\mathrm{E}(H_i^w|X_i = x)/\partial x$, which would correspond to $\partial\,\mathrm{E}(Y_{i1} - Y_{i0}|X_i = x)/\partial x$ and seems not to have attracted much attention in the treatment effect literature. Because we focus on the partial effect of X_i, we find that unlike in the treatment effect literature where the matching estimator aims to impute Y_{i1} or Y_{i0} itself, the characteristics to impute the missing potential hours in our context must be exogenous predictors of the daily hours worked.

3.2.1 Model and Estimators

To be concrete, we consider the following equation of weekly labour supply,

$$H_i^w = X_i'\beta + U_i, \qquad i = 1, \ldots, n, \tag{3.4}$$

where X_i is a $p \times 1$ vector of observable independent variables that affect hours worked with its first element being unit one. The explanatory variables X_i, including log wage in particular, tend to be correlated with U_i, and hence is often endogenous. Moreover, log wage may also be subject to measurement errors.[22] We assume that a $q \times 1$ vector of IV Z_i is available. Let $\sigma_u^2 \equiv \mathrm{Var}(U_i)$.

The ideal case is when the true weekly hours worked H_i^w were observable for each individual. The usual 2SLS estimator is then

$$\hat{\beta}_{wk} = (X'P_zX)^{-1}(X'P_zH^w), \tag{3.5}$$

where $H^w \equiv (H_1^w, \ldots, H_n^w)'$, $X \equiv (X_1, \ldots, X_n)'$, $Z \equiv (Z_1, \ldots, Z_n)'$ and $P_z \equiv Z(Z'Z)^{-1}Z'$. Since it uses the unobservable true weekly hours worked, we call it *week* estimator.

Now we consider how to utilize the ATUS hours. Because the ATUS is designed to survey about a randomly chosen day for each individual, we maintain the following assumption throughout the paper.

Assumption 1 (Random diary day)**.** *Diary day dummies* $(d_{i1}, \ldots, d_{i7})'$ *are independent from* $(X_i', Z_i', U_i', H_i^w, H_{i1}, \ldots, H_{i7})'$.

In fact, we check the correlation among the ATUS diary dummies with all the variables used in this paper, and Assumption 1 is strongly supported.

Let H^{ATUS} denote the $n \times 1$ vector of ATUS daily hours. For each $t \in \{1, \ldots, 7\}$, suppose the subsample size for diary day t is n_t, and let D_t denote an $n \times n$ diagonal matrix with elements d_{it} $(i = 1, \ldots, n)$. What D_t does is just to select the subsample for diary day t. Equation (3.3) can be re-written in such matrix notation as $H^{ATUS} = \sum_{t=1}^{7} D_t H_t$.

Since the diary day is chosen randomly, it appears natural to expect the day-to-day variation of daily hours worked within a week to cancel out in large

[22] See a survey paper by Bound, Brown and Mathiowetz (2001) for details.

samples if we pool all diary days together. This intuition leads to what we call *pool* estimator,

$$\hat{\beta}_{pool} = (X'P_zX)^{-1}(X'P_zRH^{ATUS}) = (X'P_zX)^{-1}X'P_z\Big(\sum_{t=1}^{7} r_{nt}D_tH_t\Big). \quad (3.6)$$

In eq. (3.6), $r_{nt} \equiv n/n_t$, and R is an $n \times n$ diagonal matrix with elements n/n_{t_i} $(i = 1, \ldots, n)$, which adjusts for the sampling probability of the diary days. If every day gets 1/7 probability of being sampled, then the pool estimator is equivalent to a simple 2SLS using the ATUS daily hours multiplied by seven.

The second intuitive estimator relies on the disaggregation of the weekly labour supply model into a number of daily labour supply models; that is, for $t = 1, \ldots, 7$,

$$H_{it} = X_i'\beta_t + U_{it}, \quad (3.7)$$

where $\mathrm{E}(U_{it}) = 0$. Then the parameters β in the weekly labour supply model can be re-written as $\beta = \sum_{t=1}^{7} \beta_t$. Therefore, it seems to be a logical attempt to estimate β using what we call *day* estimator, defined as

$$\hat{\beta}_{day} = \sum_{t=1}^{7} \hat{\beta}_t = \sum_{t=1}^{7} (X'P_{zt}X)^{-1}X'P_{zt}H_t, \quad (3.8)$$

where $\hat{\beta}_t$ $(t = 1, \ldots, 7)$ is the usual 2SLS estimator of β_t using only the subsample for diary day t and $P_{zt} = (D_tZ)(Z'D_tZ)^{-1}(D_tZ)'$.

Later we are going to show that both the pool estimator and the day estimator are consistent under mild conditions. However, neither of them is ideal in terms of efficiency and robustness. Instead, we propose a third feasible estimator which deviates the infeasible benchmark $\hat{\beta}_{wk}$ as little as possible, and we will show that the third estimator outperforms the first two.

In light of eq. (3.2) and the definition of P_z,[23] the infeasible 2SLS estimator

[23]Moreover, P_z is an idempotent matrix, i.e. $P_zP_z = P_z$.

$\hat{\beta}_{wk}$ can be re-written as

$$\hat{\beta}_{wk} = (X'P_zX)^{-1}X'P_z\sum_{t=1}^{7}H_t = (X'P_zX)^{-1}X'P_zZ\sum_{t=1}^{7}(Z'Z/n)^{-1}(Z'H_t/n),$$

where $H_t \equiv (H_{1t},\ldots,H_{nt})$ for each $t \in \{1,\ldots,7\}$. By the simple law of large numbers, we know that $(Z'Z/n)^{-1}(Z'H_t/n) \xrightarrow{p.} [\mathrm{E}(Z_iZ_i')]^{-1}\mathrm{E}(Z_iH_{it}')$. Assumption 1 implies that in this expression, the unconditional means equal to the conditional means, i.e. $[\mathrm{E}(Z_iZ_i')]^{-1}\mathrm{E}(Z_iH_{it}') = [\mathrm{E}(Z_iZ_i' \mid d_{it}=1)]^{-1}$ $\mathrm{E}(Z_iH_{it}' \mid d_{it}=1)$. As a result, we can use the subsample for diary day t, instead of the entire sample, to estimate the two conditional means for each t. Replace the last part of $\hat{\beta}_{wk}$ by its diary day t counterpart, we get

$$\hat{\beta}_{im} \equiv (X'P_zX)^{-1}X'P_zZ\sum_{t=1}^{7}(Z'D_tZ/n_t)^{-1}(Z'D_tH_t/n_t). \tag{3.9}$$

We call this estimator *impute* estimator. In practice, impute estimator is easy to compute using the ATUS data by the following steps:

1. ("X first stage") Regress X_i on Z_i using the entire sample and take the fitted values $\hat{X}_i$;

2. ("H first stage") For each diary day t, regress H_i^{ATUS} (i.e. H_{it_i}) on Z_i using the subsample $d_{it}=1$ to get $\hat{\alpha}_t$, and impute the weekly hours worked by $\hat{H}_i^w = \sum_{t=1}^{7}\hat{H}_{it} = \sum_{t=1}^{7}Z_i'\hat{\alpha}_t$ for the entire sample;

3. ("Second stage") Regress $\hat{H}_i^w$ on $\hat{X}_i$ using the entire sample and get $\hat{\beta}_{im}$.

Compared to the usual 2SLS estimator, this estimator adds one more simple step in the middle where the values of the unobservable weekly hours H_i^w is imputed based on the instrumental variables.

In the "H first stage", if the hours worked by individual i on day t is not observed, the impute estimator essentially matches individual i with those respondents in the diary day group t who have similar values of Z_i with her, and uses their hours worked as the imputed hours for individual i. This is similar to the "synthetic time diary" method employed by Aguiar, Bils, Charles and

Hurst (2017). It also resembles the matching estimator in the treatment effect literature, except that here we make it clear that the basis for matching has to be exogenous IV Z_i, and cannot be endogenous regressors X_i in the weekly labour supply eq. (3.4).

Remark 1 (Limitation of imputed weekly hours $\hat{H}_i^w$)**.** *It might be tempting to think of $\hat{H}_i^w$ as "predicted" weekly hours worked for worker i, and to use $\hat{H}_i^w$ to impute other variables. For example, one might propose to impute hourly wage rate by $I_i/\hat{H}_i^w$ for weekly paid workers, where I_i is weekly earning of worker i. Unfortunately, our earlier impossibility result indicates that such use of $\hat{H}_i^w$, in general, is wrong. $\hat{H}_i^w$ is merely an intermediate that facilitates efficient estimation of β. In addition, our analysis emphasizes that imputation of $\hat{H}_i^w$ should only be based on the instruments Z_i, but not endogenous X_i. Even though in many cases the latter may deliver a better "predicted" weekly hours, it results in bias in β estimates.*

Remark 2 (Exogenous X_i)**.** *If X_i are exogenous (hence X_i are their own IVs), then $\hat{\beta}_{wk} = (X'X)^{-1}(X'H^w)$ simply becomes the OLS estimator for model (3.4). It is easy to verify that in this case $\hat{\beta}_{day}$ is numerically identical to the impute estimator $\hat{\beta}_{im}$. The two differ if X_i are endogenous.*

Remark 3 (Measurement error in the ATUS hours)**.** *We acknowledge that time use surveys are not error free. Let H_{it} be the true hours worked on day t, and let $H_{it}^{ATUS} = H_{it} + e_{it}^{ATUS}$ be the ATUS hours if respondent i was interviewed for his/her hours worked on day t. Here e_{it}^{ATUS} is the measurement error of the ATUS hours. When the diary day assignment is independent of the measurement error,*[24] *and e_{it}^{ATUS} is uncorrelated with the IV Z_i, we still have $[\mathrm{E}(Z_iZ_i')]^{-1}\,\mathrm{E}(Z_iH_{it}') = [\mathrm{E}(Z_iZ_i' \,|\, d_{it}=1)]^{-1}\,\mathrm{E}(Z_iH_{it}^{ATUS\prime} \,|\, d_{it}=1)$. So the above arguments still follow.*

[24] As mentioned after Assumption 1, we found no significant correlation between the ATUS diary dummies and all the variables used in this paper, which implies that the ATUS diary day dummies are probably uncorrelated with the ATUS measurement errors.

3.2.2 Large Sample Properties

In this section, we will prove that all the feasible estimators for the ATUS that we propose are consistent under the same conditions required for the consistency of the usual 2SLS estimator. In addition, we will show that the impute estimator has superior efficiency properties. We maintain the following two assumptions throughout the paper.

Assumption 2 (Random sample)**.** *For any $i \in \{1, \ldots, n\}$, the vector $(H_i^w, H_{i1}, \ldots, H_{i7}, X_i', Z_i', d_{i1}, \ldots, d_{i7})'$ is randomly drawn from the population.*

Assumption 3 (Valid and relevant instrumental variables)**.** *Assume that $\mathrm{E}(U_i Z_i) = 0$, $\operatorname{rank} \mathrm{E}(Z_i Z_i') = q$ $(q \geq p)$, and $\operatorname{rank} \mathrm{E}(Z_i X_i') = p$.*

Theorem 1 (Consistency)**.** *Under Assumptions 1 to 3, we have that $\hat{\beta}_{wk}$, $\hat{\beta}_{im}$, $\hat{\beta}_{pool}$, and $\hat{\beta}_{day}$ all converge to β in probability as $n \to \infty$.*[25]

Remark 4 (Weak conditions for consistency of $\hat{\beta}_{day}$)**.** *We need to point out that all the estimators we consider, including the day estimator, are consistent under the weaker assumption that $\mathrm{E}(U_i Z_i) = 0$, instead of the stronger $\mathrm{E}(U_{it} Z_i) = 0$. That is, the IV only need to be valid for the weekly labour supply equation, and not necessarily so for each daily ones. Even if each daily 2SLS estimator $\hat{\beta}_t$ might be inconsistent for β_t, the day estimator $\hat{\beta}_{day}$ still is.*

The 2SLS estimator based on the CPS recalled weekly hours, on the other hand, is in general inconsistent. This is again a well known consequence of the nonclassical measurement error e_i defined in eq. (3.1).[26]

To derive the asymptotic distributions, it helps to consider the "H first stage" where the potential daily hours H_{it} are regressed on the IV Z_i:

$$H_{it} = Z_i' \alpha_t + V_{it}, \tag{3.10}$$

and let $V_t = (V_{1t}, \ldots, V_{nt})'$ denote the vector of projection residuals. By construction, $\mathrm{E}(V_{it}) = 0$ and $\mathrm{E}(Z_i V_{it}) = 0$.

[25] All proofs are in Supplementary Appendix C. Note that it is a standard result that $\hat{\beta}_{wk}$ is consistent for β and has asymptotically normal distribution.

[26] The probability limit of $\hat{\beta}_{wk}^{CPS} \equiv (X' P_z X)^{-1} (X' P_z H^{CPS})$, the 2SLS estimator based on the CPS weekly hours, is $(\mathrm{E}(X_i Z_i')[\mathrm{E}(Z_i Z_i')]^{-1} \mathrm{E}(Z_i X_i'))^{-1} \mathrm{E}(X_i Z_i')[\mathrm{E}(Z_i Z_i')]^{-1} \mathrm{E}(Z_i e_i)$, which is in general not zero, since $\mathrm{E}(Z_i e_i) \neq 0$ for the nonclassical measurement error e_i.

Assumption 4 (Diary day sampling probability)**.** *Assume that each day of a week has a positive probability of being sampled. That is,* $0 < \Pr(d_{it} = 1) < 1$ *for each day* $t \in \{1, \ldots, 7\}$.

Define $A \equiv BC^{-1}B'$ with $B \equiv \mathrm{E}(X_i Z_i')$ and $C \equiv \mathrm{E}(Z_i Z_i')$, and let $r_t = 1/\Pr(d_{it} = 1)$.

Theorem 2 (Asymptotic Normality I)**.** *Under Assumptions 1 to 4, we have the following asymptotic normality results:*

(i) $\sqrt{n}(\hat{\beta}_{wk} - \beta) \xrightarrow{d} \mathcal{N}(0, \Omega^*_{wk})$, *with* $\Omega^*_{wk} \equiv A^{-1}BC^{-1}E(U_i^2 Z_i Z_i')C^{-1}B'A^{-1}$;

(ii) $\sqrt{n}(\hat{\beta}_{im} - \beta) \xrightarrow{d} \mathcal{N}(0, \Omega^*_{im})$, *with* $\Omega^*_{im} = \Omega^*_{wk} + \Omega^*_{im-wk}$, *where*

$$\Omega^*_{im-wk} \equiv A^{-1}BC^{-1}\Big[\sum_{t=1}^{7}(r_t - 1)\,\mathrm{E}(Z_i V_{it} V_{it} Z_i') - 2\sum_{1\le t<\tau\le 7}\mathrm{E}(Z_i V_{it} V_{i\tau} Z_i')\Big]C^{-1}B'A^{-1};$$

(iii) $\sqrt{n}(\hat{\beta}_{pool} - \beta) \xrightarrow{d} \mathcal{N}(0, \Omega^*_{pool})$, *with* $\Omega^*_{pool} = \Omega^*_{im} + \Omega^*_{pool-im}$, *where*

$$\begin{aligned}&\Omega^*_{pool-im}\\ \equiv& A^{-1}BC^{-1}\Big[\sum_{t=1}^{7}(r_t - 1)\,\mathrm{E}(Z_i\alpha_t' Z_i Z_i'\alpha_t Z_i') - 2\sum_{1\le t<\tau\le 7}\mathrm{E}(Z_i\alpha_t' Z_i Z_i'\alpha_\tau' Z_i')\Big]C^{-1}B'A^{-1};\end{aligned} \tag{3.11}$$

(iv) $\sqrt{n}(\hat{\beta}_{im} - \hat{\beta}_{wk}) \xrightarrow{d} \mathcal{N}(0, \Omega^*_{im} - \Omega^*_{wk})$, *hence* $n(\hat{\beta}_{im} - \hat{\beta}_{wk})'(\Omega^*_{im} - \Omega^*_{wk})^{-1}(\hat{\beta}_{im} - \hat{\beta}_{wk}) \sim \chi^2(p)$.[27]

As is clearly shown in the proof, both Ω^*_{im-wk} and $\Omega^*_{pool-im}$ are variance-covariance matrices of some random vectors (i.e. positive definite), which in

[27] Here we do not provide the asymptotic variance for $\hat{\beta}_{day}$, and there are two reasons. (1) As explained in Remark 8 below, none of the asymptotic variance formulas given here is estimable using the ATUS data under Assumptions 2 to 4. (But they will be estimable if Assumption 5 below holds.) (2) Assumption 3 only guarantees that the IVs are valid for the weekly labour supply equation, but not necessarily for the daily ones, so $\hat{\beta}_t$ might be inconsistent for β_t for some t. The asymptotic distribution of IV estimators, when the IVs are invalid, is very complicated in general (for details, see Kiviet and Niemczyk, 2009).

turn implies that $\Omega^*_{pool} \geq \Omega^*_{im} \geq \Omega^*_{wk}$. Although the pool estimator is intuitive and consistent, the impute estimator is better in terms of efficiency. In general, $\hat{\beta}_{day}$ does not follow a fixed efficiency ranking compared with $\hat{\beta}_{im}$.

Remark 5 (Relative efficiency of $\hat{\beta}_{wk}$)**.** *It is not surprising that the infeasible estimator $\hat{\beta}_{wk}$ is the most efficient should the true weekly hours worked were observed. The efficiency difference between $\hat{\beta}_{wk}$ and $\hat{\beta}_{im}$ results from the fact that $\hat{\beta}_{im}$ only utilizes diary day subsamples to impute $\hat{H}_{it}$ (and sum $\hat{H}_{it}$ to get $\hat{H}^w_i$), while $\hat{\beta}_{wk}$ directly imputes $\hat{H}^w_i$ using the entire sample.*[28]

Remark 6 (Relative efficiency of $\hat{\beta}_{im}$)**.** *The efficiency gain of $\hat{\beta}_{im}$ compared to $\hat{\beta}_{pool}$ might be less expected. But is also very intuitive—to impute $\hat{H}_{it}$, $\hat{\beta}_{im}$ uses only data on H_{it}, the relevant diary day observations. On the contrary, $\hat{\beta}_{pool}$ uses data on both H_{it} and $H_{i\tau}$ ($\tau \neq t$), and $H_{i\tau}$ observations ($t \neq \tau$) merely add noise, which results in a less efficient estimator. A less obvious point is that the size of the efficiency gap depends on the diary day sampling weights. In the extreme case where there is no variation in daily hours (i.e. $H_{i1} = \cdots = H_{i7}$, and hence $\mathrm{E}(Z_i V_{it} V_{it} Z_i') = \mathrm{E}(Z_i V_{it} V_{i\tau} Z_i')$ and $\mathrm{E}(Z_i \alpha_t' Z_i Z_i' \alpha_t Z_i') = \mathrm{E}(Z_i \alpha_t' Z_i Z_i' \alpha_\tau Z_i')$ for all $t, \tau = 1, \ldots, 7$), one might think that it does not matter which day gets surveyed, and hence the pool estimator (subject to sampling weights adjustment) suffices. However, part (iii) of Theorem 2 shows that for $\Omega^*_{pool-im}$ to be zero, we need equal sampling weights so that $r_t = \lim_{n\to\infty} n/n_t = 7$ for $t = 1, \ldots, 7$. Otherwise $\Omega^*_{pool} > \Omega^*_{im} > \Omega^*_{wk}$ remains. This means that, given the sampling weights of the ATUS diary days (i.e. $r_1 = r_7 = 4$ and $r_2 = \cdots r_6 = 10$), the impute estimator will be more efficient than the pool estimator even when there is no variation in daily hours.*

Remark 7 (Hausman test between the CPS and the ATUS)**.** *Part (iv) of Theorem 2 indicates that we can test the presence of nonclassical measurement errors in the recalled weekly hours in the CPS using the Hausman test. Under the null hypothesis of no nonclassical measurement errors, the 2SLS based on the recalled weekly hours in the CPS will be consistent and as efficient as the week estimator $\hat{\beta}_{wk}$; while under the alternative, such 2SLS will be biased. In*

[28] Note that $\hat{\beta}_{wk} = (X'P_zX)^{-1}(X'P_zH^w) = (X'P_zP_zX)^{-1}(X'P_zP_zH^w) = (\hat{X}'\hat{X})^{-1}(\hat{X}'\hat{H}^w)$ since P_z is an idempotent matrix.

both cases, the impute estimator $\hat{\beta}_{im}$ is consistent but less efficient.

Remark 8 (Standard errors of $\hat{\beta}_{wk}$, $\hat{\beta}_{im}$ and $\hat{\beta}_{pool}$)**.** *Even though Theorem 2 gives explicit formulas for the asymptotic variances and provides profound insights to the relative efficiency of the estimators, the standard errors cannot be computed based on Theorem 2 using the ATUS data alone because none of Ω^*_{wk}, Ω^*_{im} or Ω^*_{pool} is estimable without further assumptions. To see this, note that $E(U_i^2 Z_i Z_i')$, $E(Z_i V_{it} V_{i\tau} Z_i')$ and $E(Z_i \alpha_t' Z_i Z_i' \alpha_\tau Z_i')$ $(1 \leq t < \tau \leq 7)$ require observing the same individual on different days, but the ATUS only provides information on one single day for each individual, and hence is inadequate in this regard. This is just another manifestation of the impossibility results discussed in Section 3.1.*

In order to compute the standard errors of the estimators, two approaches are feasible. For the empirical results in Section 4.2, Section 5 and Supplementary Appendix B, we will report the standard errors computed using both approaches, and they happen to be very close to each other.

1. One may first estimate $E(U_i^2 Z_i Z_i')$, $E(Z_i V_{it} V_{i\tau} Z_i')$ and $E(Z_i \alpha_t' Z_i Z_i' \alpha_\tau Z_i')$ $(1 \leq t < \tau \leq 7)$ with the help of other data sources that observe one individual on multiple days, and then plug those estimates into the formulas provided in Theorem 2.

   (a) The ideal data source is a pilot survey that observes a smaller but representative sample of individuals for multiple days, which could be conducted alongside the ATUS.[29]

   (b) Without such pilot survey, an alternative source is the DTUS, if one is willing to assume that the correlations among daily hours in the DTUS are the same as those in the ATUS. In Supplementary Appendix A, we describe in details how to compute the standard errors using this approach.

2. Another approach results from the observation that if one is willing to make the following slightly stronger assumption about the instrumental

[29] In fact, we recommend such pilot survey to be conducted alongside the ATUS in order to obtain the standard errors.

variables, then the asymptotic variances no longer depend on the correlations among daily hours on multiple days. The standard error formulas will be given in Theorem 4.

Assumption 5 (Instrumental variable in daily equations)**.** *Assume that* $\mathrm{E}(U_{it}Z_i) = 0$ *for all* $t = 1, \ldots, 7$*, that is, the instrumental variables are valid in the daily labour supply equations.*

To state the results, it helps to consider the "X first stage" where the (potentially endogenous) regressors X_i are regressed on the IV Z_i:

$$X_i = \lambda Z_i + \epsilon_i, \tag{3.12}$$

and we have $\mathrm{E}(\epsilon_{it}) = 0$ and $\mathrm{E}(Z_i\epsilon_i) = 0$ by construction.

Theorem 3 (Asymptotic Normality II)**.** *Under Assumptions 1 to 5, we have the following asymptotic normality results:*

(i) $\sqrt{n}(\hat{\beta}_{im} - \beta) \xrightarrow{d.} \mathcal{N}(0, \Omega_{im})$*, with*

$$\Omega_{im} \equiv \Omega_{im,1} + \Omega_{im,2} + 2\Omega_{im,3}, \text{ where} \tag{3.13}$$

$$\begin{aligned}
\Omega_{im,1} \equiv & A^{-1}BC^{-1}\Big[\sum_{t=1}^{7}(r_t - 1)\,\mathrm{E}(Z_i\beta_t'\epsilon_i\epsilon_i'\beta_t Z_i') - 2\sum_{1\le t<\tau\le 7}\mathrm{E}(Z_i\beta_t\epsilon_i'\epsilon_i\beta_\tau Z_i')\Big]C^{-1}B'A^{-1},\\
\Omega_{im,2} \equiv & A^{-1}BC^{-1}\Big[\sum_{t=1}^{7} r_t\,\mathrm{E}\big(U_{it}^2 Z_i Z_i'\big)\Big]C^{-1}B'A^{-1},\\
\Omega_{im,3} \equiv & A^{-1}BC^{-1}\Big[\sum_{t=1}^{7}(r_t - 1)\,\mathrm{E}(U_{it}Z_i\beta_t'\epsilon_i Z_i') - 2\sum_{1\le t<\tau\le 7}\mathrm{E}(U_{i\tau}Z_i\beta_t'\epsilon_i Z_i')\Big]C^{-1}B'A^{-1};
\end{aligned}$$

(ii) $\sqrt{n}(\hat{\beta}_{pool} - \beta) \xrightarrow{d.} \mathcal{N}(0, \Omega_{pool})$*, with* $\Omega_{pool} \equiv \Omega_{im} + \Omega_{pool-im}$*, where* Ω_{im} *is defined in eq.* (3.13) *and* $\Omega_{pool-im} = \Omega^*_{pool-im}$ *is defined in eq.* (3.11)*;*

(iii) $\sqrt{n}(\hat{\beta}_{day} - \beta) \xrightarrow{d.} \mathcal{N}(0, \Omega_{day})$*, with*

$$\Omega_{day} \equiv A^{-1}BC^{-1}\Big[\sum_{t=1}^{7} r_t\,\mathrm{E}(U_{it}^2 Z_i Z_i')\Big]C^{-1}B'A^{-1}.$$

The asymptotic variance formulas in Theorem 3 lead to easy-to-compute standard errors for $\hat{\beta}_{im}$, $\hat{\beta}_{pool}$ and $\hat{\beta}_{day}$. Before giving the standard error formulas, we need some notation.

Note that the coefficient λ in the "X first stage" eq. (3.12) can be estimated using the entire sample. For each individual $i = 1, \ldots, n$, let $\hat{X}_i \equiv Z_i'\hat{\lambda}$ and $\hat{\epsilon}_i \equiv Z_i - \hat{X}_i$. Let $A_n \equiv \frac{1}{n}\sum_{i=1}^n \hat{X}_i\hat{X}_i'$, $B_n \equiv \frac{1}{n}\sum_{i=1}^n X_iZ_i'$ and $C_n \equiv \frac{1}{n}\sum_{i=1}^n Z_iZ_i'$. Note that the coefficients α_t in the "H first stage" eq. (3.10) can be estimated using the subsample for diary day t. For each individual $i = 1, \ldots, n$, let $\hat{U}_{it_i} \equiv H_{it_i} - Z_i\hat{\beta}_{t_i}$. Recall that t_i denote the diary day for individual i in the ATUS data. For each $t = 1, \ldots, 7$, define

$$\hat{Q}_t^{uz} \equiv \frac{1}{n_t}\sum_{i:t_i=t} \hat{U}_{it_i}^2 Z_iZ_i', \qquad \hat{Q}_t^{\alpha} \equiv \frac{1}{n}\sum_{i=1}^n Z_i\hat{\alpha}_t'Z_iZ_i'\hat{\alpha}_tZ_i',$$

$$\hat{Q}_t^{z\epsilon} \equiv \frac{1}{n}\sum_{i=1}^n Z_i\hat{\beta}_t'\hat{\epsilon}_i\hat{\epsilon}_i'\hat{\beta}_tZ_i', \qquad \hat{Q}_t^{u\epsilon} \equiv \frac{1}{n_t}\sum_{i:t_i=t} \hat{U}_{it_i}Z_i\hat{\beta}_t'\hat{\epsilon}_iZ_i'.$$

And for each pair of (t, τ) such that $1 \leq t < \tau \leq 7$, define

$$\hat{Q}_{t,\tau}^{z\epsilon} \equiv \frac{1}{n}\sum_{i=1}^n Z_i\hat{\beta}_t'\hat{\epsilon}_i\hat{\epsilon}_i'\hat{\beta}_\tau Z_i',\ \hat{Q}_{t,\tau}^{u\epsilon} \equiv \frac{1}{n_\tau}\sum_{i:t_i=\tau} \hat{U}_{it_i}Z_i\hat{\beta}_t'\hat{\epsilon}_iZ_i',\ \hat{Q}_{t,\tau}^{\alpha} \equiv \frac{1}{n}\sum_{i=1}^n Z_i\hat{\alpha}_t'Z_iZ_i'\hat{\alpha}_\tau Z_i'.$$

Therefore, we can define

$$\begin{aligned}
\hat{\Omega}_{im,1} \equiv& A_n^{-1}B_nC_n^{-1}\Big[\sum_{t=1}^7 (r_{nt}-1)\hat{Q}_t^{z\epsilon} - 2\sum_{1\leq t<\tau\leq 7}\hat{Q}_{t,\tau}^{z\epsilon}\Big]C_n^{-1}B_n'A_n^{-1},\\
\hat{\Omega}_{im,2} \equiv& A_n^{-1}B_nC_n^{-1}\Big[\sum_{t=1}^7 r_{nt}\hat{Q}_t^{uz}\Big]C_n^{-1}B_n'A_n^{-1},\\
\hat{\Omega}_{im,3} \equiv& A_n^{-1}B_nC_n^{-1}\Big[\sum_{t=1}^7 (r_{nt}-1)\hat{Q}_t^{u\epsilon} - 2\sum_{1\leq t<\tau\leq 7}\hat{Q}_{t,\tau}^{u\epsilon}\Big]C_n^{-1}B_n'A_n^{-1},\\
\hat{\Omega}_{pool-im} \equiv& A_n^{-1}B_nC_n^{-1}\Big[\sum_{t=1}^7 (r_{nt}-1)\hat{Q}_t^{\alpha} - 2\sum_{1\leq t<\tau\leq 7}\hat{Q}_{t,\tau}^{\alpha}\Big]C_n^{-1}B_n'A_n^{-1}, \text{ and}
\end{aligned}$$

$$\hat{\Omega}_{day} \equiv A_n^{-1} B_n C_n^{-1} \Big[\sum_{t=1}^{7} r_t \hat{Q}_t^{uz} \Big] C_n^{-1} B_n' A_n^{-1}.$$

Finally, define $\hat{\Omega}_{im} = \hat{\Omega}_{im,1} + \hat{\Omega}_{im,2} + \hat{\Omega}_{im,3}$ and $\hat{\Omega}_{pool} = \hat{\Omega}_{im} + \hat{\Omega}_{pool-im}$.

Theorem 4 (Standard errors)**.** *Under Assumptions 1 to 5, we have the following results: (i)* $\hat{\Omega}_{im} \xrightarrow{p.} \Omega_{im}$*; (ii)* $\hat{\Omega}_{pool} \xrightarrow{p.} \Omega_{pool}$*; (iii)* $\hat{\Omega}_{day} \xrightarrow{p.} \Omega_{day}$*.*

4 Lessons from the Dutch Time Use Survey

The sample we use in this section consists of individuals from the DTUS (Fisher et al., 2018) aged between 25 and 54 surveyed in 1980, 1985, 1990, 1995, 2000 and 2005, whose recalled hours and diary hours are both positive. The entire sample contains 6,567 individual-year records.

4.1 Simulations

Based on the DTUS data, we design a simulation study to compare the finite sample performance of the estimators discussed previously. The nice thing about the DTUS is that it contains CPS-type recalled weekly hours, as well as daily diary hours for an entire week. As a result, we are able to compute the week estimator $\hat{\beta}_{wk}$, which would have been impossible for the ATUS.

Given the daily hours worked H_{it}^{DTUS} $(t = 1, \ldots, 7)$ in the DTUS, we generate a single endogenous regressor $\tilde{X}_i$ and a single instrumental variable $\tilde{Z}_i$ such that eq. (3.7) is satisfied with $X_i = (1, \tilde{X}_i)'$, $Z_i = (1, \tilde{Z}_i)'$, $\text{Corr}(U_{it}, Z_i) = 0$ for $t = 1, \ldots, 7$. In particular, let H^{DTUS} denote the $n \times 7$ matrix with elements H_{it}^{DTUS}, and let $T_1, \ldots, T_7$ be the principal components of H^{DTUS}. We set $\tilde{Z}_i$ to be the first principal component of H^{DTUS}, i.e. $\tilde{Z} = T_1$. To introduce the endogeneity, we generate an $n \times 7$ matrix of independent random variables from $N(0, 2)$,[30] denoted by V. Then we set $H_{it} = H_{it}^{DTUS} + V_{it}$ and $\tilde{X}_i = \tilde{Z}_i + \rho \sum_{t=1}^{7} V_{it}$ for $i = 1, \ldots, n$ and $t = 1, \ldots 7$. The true parameters β_t are therefore just the weights in H_t $(t = 1, \ldots, 7)$ associated with the first

[30] Such that $\text{Var}(U_i) \approx \text{Var}(T_1)$ in the exogenous regressor case.

principal component. The true value of β in eq. (3.4) is 2.2694.[31] By varying ρ, we vary $\text{Corr}(\tilde{X}_i, U_i)$, the degree of endogeneity of the regressor $\tilde{X}_i$. When $\rho = 0$, the regressor is exogenous, and we try other values of ρ such that $\text{Corr}(\tilde{X}_i, U_i) \in \{0, 0.25, 0.5, 0.75\}$. Note that as ρ increases, the strength of the IV also decreases. For the above values of ρ, $\text{Corr}(\tilde{X}_i, \tilde{Z}_i)$ equals 1, 0.95, 0.80 and 0.43, respectively.

To evaluate the finite sample performance of the various estimators considered in Section 3, we randomly draw a subsample of size $n \in \{250, 500, 1000, 2500\}$. Then we generate fictitious ATUS-type samples by randomly choosing only one day for each individual in the drawn subsamples using the diary day sampling weights of the ATUS. [32] We repeat the experiment $10,000$ times, and Table 1 reports the mean squared errors (MSE), squared biases and variances for all estimators considered.

Some patterns are apparent. First, the usual 2SLS estimator using the CPS-type recalled weekly hours, $\hat{\beta}_{re}$, has the largest MSE in almost all parameterizations, which is roughly ten times larger than the maximum among all the other estimators. The large MSE is nearly entirely driven by the large bias, which is in turn a result of nonclassical measurement error in the CPS-type recalled weekly hours. Below we will illustrate this nonclassical measurement error using the DTUS data in Figure 2. Second, for almost all parameterizations, the biases of all the estimators based on the diary hours are negligible, and the differences in the performance of $\hat{\beta}_{wk}$, $\hat{\beta}_{im}$, $\hat{\beta}_{pool}$ and $\hat{\beta}_{day}$ reside in efficiency and robustness. Third, since the infeasible week estimator $\hat{\beta}_{wk}$ uses the diaries of an entire week, it is much more efficient than the others. This verifies the result of Theorem 2. Fourth, the impute estimator $\hat{\beta}_{im}$ is more efficient than $\hat{\beta}_{pool}$ and $\hat{\beta}_{day}$ in all parameterizations. Again, this verifies the result of Theorem 2. Fifth, when the regressor is exogenous, $\hat{\beta}_{im}$ and $\hat{\beta}_{day}$ perform equally well. This is because, as we mentioned before, the two estimators are numerically the same in this case. Last but not least, the day estimator $\hat{\beta}_{day}$ appears to be unstable, especially when the sample size is smaller and when

[31] $\beta_1 = 0.0007$, $\beta_2 = 0.4379$, $\beta_3 = 0.4554$, $\beta_4 = 0.4576$, $\beta_5 = 0.4528$, $\beta_6 = 0.4304$ and $\beta_7 = 0.0346$.

[32] That is, the probability of being drawn is 0.25 for $t = 1$ (Sundays) and $t = 7$ (Saturdays), and 0.1 for the others.

the IV is weaker. The reason is that $\hat{\beta}_{day}$ relies on the daily 2SLS estimators $\hat{\beta}_t$. When the sample size is small, the effective sample size for each day gets even smaller, and taking the inverse of the sample average matrices magnifies the sampling errors substantially.[33]

4.2 Labour Supply Elasticity Estimates

In this section, we illustrate the empirical impacts on the labour supply elasticity estimates of both nonclassical measurement errors and time specificity using the DTUS.

Figure 2 shows the measurement error in the recalled weekly hours worked in the Dutch data. The "measurement error" in Figure 2 equals the recalled weekly hours worked minus the weekly hours worked from the seven-day diaries in the DTUS.[34] If the recalled hours worked do not have nonclassical measurement error, then the measurement error in Figure 2 would be uncorrelated with the weekly hours from the seven-day diaries. Panel A of Figure 2 suggests the opposite: the measurement error in the recalled hours is negatively correlated with the hours from time use survey. Its kernel density (panel B of Figure 2) suggests that more people overstate the recalled hours worked then understate. The negative correlation between the measurement error and the true hours worked coincide with the observation made by Bound et al. (1989) about the PSID.

We estimate the labour supply elasticities using the following model,

$$H_i^w = \beta_0 + \beta_1 kids_i + \beta_2 edu_i + \beta_3' X_i + U_i, \tag{4.1}$$

where $kids_i$ is the number of kids aged under 18, edu_i includes two dummy variables, one for completing secondary education and the other for higher

[33]We also conduct the same simulations based only on the five weekdays in the DTUS. The results are qualitatively the same and are reported in Table B.2 in Appendix B.

[34]In the DTUS, the recalled weekly hours combine three questions in the survey: (1) hours worked in the previous week; (2) usual weekly hours worked in the previous year; or (3) the seven-day diary hours. The answer to the next question will be used if the respondents are unable to answer the previous question(s), but it is not indicated the answer to which question was actually used for each individual. Probably due to this, many respondents in the DTUS exhibit "zero measurement errors" in the recalled weekly hours.

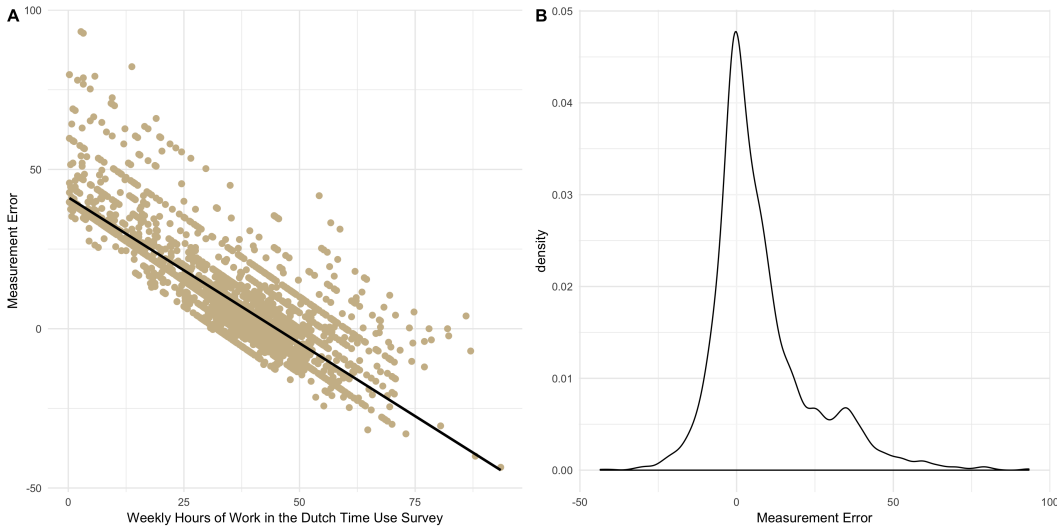

Figure 2: Measurement Errors in the DTUS Recalled Weekly Hours Worked

Panel A (left): scatter plot of the measurement errors in recalled weekly hours worked v.s. the DTUS weekly hours worked. Panel B (right): kernel density of the measurement errors. In both, the measurement errors are the differences between the recalled weekly hours worked subtracting the DTUS weekly hours worked for the same individuals.

than secondary education, and X_i is a vector of control variables, including age, age-squared, a dummy of working in private sector, an urban area dummy, and year dummies.

Table 2 shows the effects of the number of children and education on labour supply. We used both the recalled weekly hours worked ($\hat{\beta}_{re}$) and the seven-day diary hours ($\hat{\beta}_{wk}$) as the dependent variable. We also randomly draw one day for each respondent, then apply our impute estimator ($\hat{\beta}_{im}$). For both married men and married women, $\hat{\beta}_{re}$ are considerably different from $\hat{\beta}_{wk}$ and $\hat{\beta}_{im}$, with different signs when significant. In the meantime, the latter two always have the same signs, even though the magnitudes may differ. We conduct joint Hausman tests for the three coefficients in the table between $\hat{\beta}_{re}$ and $\hat{\beta}_{im}$, and between $\hat{\beta}_{wk}$ and $\hat{\beta}_{im}$. For both married women and married men, the Hausman tests reject the null hypotheses $\hat{\beta}_{re} = \hat{\beta}_{im}$ but do not reject $\hat{\beta}_{wk} = \hat{\beta}_{im}$.

Based on the time use survey hours, both $\hat{\beta}_{wk}$ and $\hat{\beta}_{im}$ indicate that the effects on married women's labour supply are significantly negative for more children and significantly positive for higher education. The recalled hours, on

the other hand, produce $\hat{\beta}_{re}$ estimates that are too noisy to draw a conclusion.

5 Comparing Labour Supply Elasticity Estimates Using the ATUS and the CPS

In this section, we compare the labour supply elasticity estimates resulting the CPS recalled weekly hours and the ATUS daily diary hours.

5.1 Empirical Sample and Summary Statistics

The data are from the 2003–2017 ATUS (Hofferth et al., 2018). As mentioned in Section 2, the ATUS sample is randomly drawn from the outgoing rotation group of the CPS respondents.[35] Therefore, for every respondent in the ATUS, we have their answers to all CPS questions as well. The sample used for our empirical analysis consists of hourly paid workers[36] aged between of 25 and 54, whose wage rate is positive, and spouse earnings (if married) and total usual weekly hours worked at current job reported in the last CPS interview are observed. The age restriction is to avoid complications of schooling and retirement decisions. The hourly wage rate was trimmed at percentiles 1 from below and 99 from above. After the trimming, the hourly wage in the sample ranges from \$5.2 to \$67.8 for men and from \$3.6 to \$63.1 for women (2017 US dollars).

We argue that the discrepancies between the ATUS sample and the CPS sample are small, and the reasons are as follows. First, if the respondent changed job (or changed employment status) since the last CPS interview, then her answers to related CPS questions are updated at the time of the ATUS interview, and we use the updated CPS hours whenever applied. This eliminates the discrepancy due to job or employment status change. Second, we include only the respondents who answered both the CPS and the ATUS

[35] The ATUS is conducted, on average, two to five months after their participation in the CPS.

[36] We exclude salaried workers because their hourly wage rate is much harder to measure. In a typical survey, the hourly wage for salaried workers is total earnings during a particular period divided by the hours worked in that period.

questions for themselves. This removes the discrepancies due to someone speculating someone else's CPS hours.[37] Third, we verify that those respondents who made their way into the ATUS sample are representative of the larger CPS sample,[38] even though the response rate of the ATUS might seem low to sharp eyes.[39]

Panel A of Table 3 provides means and standard deviations of the hours worked and hourly wage rate, computed using both the CPS and the ATUS for the same respondents in our empirical analysis sample. The CPS weekly hours worked we use is the number of hours per week that the respondent usually works at his/her current job at the reported hourly wage rate.[40] Here we calculate a lower bound in the following way. It is reasonable to assume that the correlation between the hours worked by the same person in two days, H_{it} and $H_{it'}$, is nonnegative. By $H_i^w = \sum_{t=1}^{7} H_{it}$, we have $\mathrm{Var}(H_i^w) \geq \sum_{t=1}^{7} \mathrm{Var}(H_{it})$, where $\mathrm{Var}(H_{it})$ can be readily estimated by the sample variance of hours worked on day t in the ATUS. According to Table 3, men work slightly more hours than women regardless of marital status and data source; married men work slightly more hours than unmarried men, but married women work less than unmarried women; and for both genders, the married have higher hourly wage rates than the unmarried.

[37]Every respondent in the ATUS records time diary for themselves, but the household head might answer the CPS questions on behalf of other household members.

[38]We compare our empirical analysis sample (from the ATUS) to two larger CPS samples. The first is the 2003–2017 CPS sample (regardless of whether the respondent took part in the ATUS or not) after applying the same criterion (age, trimming of hourly wage, etc.) used for our empirical analysis sample, and the other is the entire 2003-2017 CPS sample of hourly paid workers (all age, no trimming of hourly wage, etc.). In Supplementary Appendix B (submitted together with this paper), Table B.4 tabulates the summary statistics of many key variables for all these three samples, and all of the them are essentially the same across the three samples. Moreover, Table B.5 in Supplementary Appendix B reports the summary statistics of weekly hours and the weekly labour supply elasticity estimates based on the first larger CPS sample, and they are very close to the CPS based estimates reported in Table 3 of this paper.

[39]The average response rate of the ATUS is roughly 50%, and that of the CPS is higher than 80%.

[40]By the potential outcome argument, the standard deviation of the ATUS imputed weekly hours worked is impossible to compute without ad hoc assumptions.

5.2 Labour Supply Elasticities

We estimate the labour supply elasticities using the following linear regression model,

$$H_i^w = \beta_0 + \beta_1 \ln w_i + \beta_2 y_i^{sp} + \beta_3' X_i + U_i, \tag{5.1}$$

where $\ln w_i$ is the natural log of hourly wage, y_i^{sp} is the usual weekly earnings of i's spouse ($y_i^{sp} = 0$ for unmarried worker), and X_i is a vector of control variables, including age, age-squared, the number of children aged below 5, the number of children aged between 5 and 18, two education dummies, eight Census division dummies, a metropolitan area dummy, race dummies, year dummies, occupation dummies and industry dummies. For hourly paid workers, ATUS directly asks them the hourly wage rate. Note that our sample only consists of those individuals who participated in both the ATUS and the CPS. As a result, despite the use of different measures of hours worked, $\hat{\beta}_{re}$ and $\hat{\beta}_{im}$ are built on the same sample.

In order to safeguard against the potential classical measurement error problem in wage and spouse weekly earnings, we use wage deciles and spouse earning deciles as IV.[41] The reported estimates of elasticities here are evaluated at the respective sample mean hours worked per week.

Panel B of Table 3 shows the estimation results by gender and marital status. For each explanatory variable of interest, we report both $\hat{\beta}_{re}$, which is based on the CPS recalled weekly hours and $\hat{\beta}_{im}$, which is based on the ATUS daily hours and our proposed imputation method. For the CPS based $\hat{\beta}_{re}$, the standard errors are just the usual 2SLS standard errors. For the ATUS based $\hat{\beta}_{im}$, however, we report both standard errors computed using the two approaches discussed in Section 3.2.2. The standard errors using the formulas in Theorem 4 under strong assumptions about IV (Assumption 5) are reported in round parentheses, and the ones using the correlation coefficients from the DTUS are reported in square parentheses. The two approaches happen to

[41] We report OLS estimates in Table B.10 of the online Appendix, and the results are almost the same, both qualitatively and quantitatively. Our choice of IV follows the suggestion by Juhn and Murphy (1997) and Blau and Kahn (2007). The reason that wage (or spouse earning) decile serves as a valid IV for wage (or spouse earning) in the presence of classical measurement errors is that we believe that the variation in the measurement error is not big enough to alter the decile grouping for a substantial proportion of respondents.

produce standard errors that are very close to each other.[42] We conduct joint Hausman tests of the coefficients of the variables appearing in Panel B between the CPS and the ATUS. The p-values are smaller than 0.1 for unmarried men and married women.[43]

Both the CPS and the ATUS indicate that women's labour supply is more wage-elastic than that of men, with the labour supply of married women having the largest wage elasticity (0.1589 and 0.1048 respectively). Compared to the CPS, the ATUS results in smaller own wage elasticities across the board, and this agrees with what Barrett and Hamermesh (2019) found.[44] From the CPS to the ATUS, the reduction of own wage elasticities for men exceeds that for women, raising the relative own wage elasticities for women.[45] For married women, the ATUS yields much smaller cross earning elasticity than implied by the CPS (-0.0579 v.s. -0.0943). For married men, the CPS indicates that their labour supply is non-elastic with respect to spouse earnings (-0.0019), consistent with previous findings in the literature (for example, Blau and Kahn, 2007); notwithstanding, the ATUS produces a much higher cross earning elasticity, and it is comparable with that of married women (-0.0347).[46]

Our estimates based on the CPS are on par with those in the literature.

[42]This by no means indicates that the correlations among daily hours in the DTUS are close to those in the ATUS. In fact, since the formulas in Theorem 4 do not rely on the correlations, they are consistent with any correlation patterns in the ATUS. The fact that we had to use the DTUS correlations to implement the formulas in Theorem 2 highlights our suggestion of a pilot survey (detailed in Section 6).

[43]The empirical findings in Table 3 are very robust to choice of IVs, subsamples, or definition of "work" activities. Robustness checks are provided in Supplementary Appendix B.

[44]Note that this pattern cannot be explained by respondents bunching their recalled weekly hours at 40, the usual suspect of nonclassical measurement errors, which alone will result in lower elasticities from the CPS than from the ATUS.

[45]Heckman (1993) argues that an important reason that married women display higher own wage elasticity than men and unmarried women is because their labour force participation decision is more wage-elastic, and that basing the estimation only on those who work essentially compares married women's higher extensive margin elasticity with other groups' intensive margin elasticities. Due to the lack of good instruments, we don't correct for the sample selection bias and acknowledge that our elasticity estimates are hybrid of both margins.

[46]The Hausman test of this single coefficient rejects the null hypothesis of equal coefficients between the CPS and the ATUS.

Across roughly twenty estimates surveyed by Blundell and MaCurdy (1999), the median own wage labour supply elasticity is 0.08 for men and 0.78 for married women. For cross wage elasticities and conditional on having positive hours, Devereux (2004) reports around -0.06 for men and around -0.5 for women in the 1980s. For married women, Blau and Kahn (2007) document robust and substantial decline in married women's labour supply elasticities from 1980s to 2000s. Their own wage elasticity fell from roughly 0.77 in 1980 to roughly 0.36 in 2000. Their cross wage elasticity decreased from around -0.33 in 1980 to around -0.19 in 2000. Since our sample covers the years 2003 - 2017, the fact that all of our CPS based estimates have smaller absolute values than those for 2000 in Blau and Kahn (2007) is consistent with the decline in the responsiveness of married women's labour supply.[47]

Panel B of Table 3 also give interesting elasticity estimates with respect to number of kids. For married women, both surveys lead to very large and almost identical elasticities with respect to number of younger kids for married women (-0.0897 and -0.0858); and yet the ATUS yields a much smaller elasticity with respect to number of older kids than the CPS (-0.012 v.s. -0.0287). For married men, the ATUS implies more elastic labour supply with respect to numbers of kids than the CPS as well.

It is worth mentioning that there are many possible sources that result in different elasticity estimates between the CPS and the ATUS (see Section 5 of Bound, Brown and Mathiowetz, 2001, for example), and the mean-reverting error[48] is only one of them. In fact, if the mean-reverting error was the only reason for different estimates between the CPS and the ATUS, then all the ATUS elasticity estimates (with respect to own wage, spouse earning, and number of kids) would all have had larger absolute values than their CPS counterparts. This contradicts our empirical findings in Table 3, where the ATUS indicates more elastic labour supply with respect to some regressors for some groups and the opposite in other cases. The same patterns are very

[47] In addition, Blau and Kahn (2007) report men's own wage elasticities around 0.1 without notable time trend.

[48] That is, people who work more hours tend to under-report, and those who work fewer hours tend to over-report.

robust across various robustness checks we conduct.[49]

6 Comments on Time Use Survey Design

For $\hat{\beta}_{im}$ to have the same precision as $\hat{\beta}_{wk}$, how much larger the sample size have to be? To get a rough idea, let's assume homoskedasticity so that Ω^*_{im-wk} and Ω^*_{wk} simplify to $\Omega^*_{im-wk} = \left[\sum_{t=1}^{7}(r_t-1)\,\mathrm{E}(V_{it}^2)-2\sum_{1\le t<\tau\le 7}\mathrm{E}(V_{it}V_{i\tau})\right]A^{-1}$ and $\Omega^*_{wk} = \mathrm{E}(U_i^2)A^{-1}$. Using the DTUS data[50] and we get $\hat{\mathrm{E}}(U_i^2) = 146.2$ and $\sum_{t=1}^{7}(r_t-1)\hat{\mathrm{E}}(V_{it}^2) - 2\sum_{1\le t<\tau\le 7}\hat{\mathrm{E}}(V_{it}V_{i\tau}) = 409.1$. Hence the estimates of the asymptotic variances of $\hat{\beta}_{wk}$ and $\hat{\beta}_{im}$ are $\widehat{\mathrm{Var}}(\hat{\beta}_{wk}) = n^{-1}146.2A^{-1}$ and $\widehat{\mathrm{Var}}(\hat{\beta}_{im}) = n^{-1}(146.2+409.1)A^{-1}$.

If the correlation coefficients among the impute residuals of hours worked across different days in the ATUS are the same as in the DTUS, then such back-of-envelop calculation implies that compared to a survey that records the respondents' activities for an entire week and enables the use of the week estimator $\hat{\beta}_{wk}$, the number of respondents surveyed in the ATUS has to be roughly 3.8 times in order to get an impute estimator $\hat{\beta}_{im}$ with the same precision.

If the average costs of following the same individuals for seven consecutive days is higher than 3.8 times of following them for one day, although the latter is justified from efficiency point of view, it is still worthwhile to do the former, at least in a smaller pilot sample. The reason is that we need the correlation coefficients among the impute residuals of hours worked across different days to compute the standard error of the impute estimator $\hat{\beta}_{im}$, as provided in Theorem 2. Besides, the time use survey hours become less reliable as the period of survey gets longer.

It is apparent that Ω_{im-wk}, the measure of relative inefficiency of $\hat{\beta}_{im}$ compared to $\hat{\beta}_{wk}$, depends on the diary day sampling probabilities $1/r_t$ ($t = 1,\ldots,7$). If efficiency of $\hat{\beta}_{im}$ is our primary concern, then we can minimize Ω_{im-wk} by choosing r_t subject to the constraint $\sum_{t=1}^{7}\frac{1}{r_t} = 1$. The optimal

[49] Reported in Table B.8 to Table B.11 in Supplementary Appendix B.

[50] The variables in Z_i are the same as in Section 4.2, and the diary day sampling weights are in accordance with the ATUS weights.

sampling probabilities are $1/r_t = \frac{\sigma_t^2}{\sum_{s=1}^{7} \sigma_s^2}$; that is, more weights should be given to the days on which the hours worked exhibit larger variation among the population.

7 Conclusion

In this paper, we use the familiar potential outcome framework to demonstrate that weekly hours worked (or their distribution function) cannot be recovered from typical time use surveys. In spite of this impossibility result, important parameters of labour supply can still be consistently and relatively efficiently estimated using time use surveys. We discuss the large sample properties of several intuitive estimators and recommend the impute estimator on the ground of efficiency and robustness. The impute estimator is a simple modification of the usual 2SLS estimator, which imputes the dependent variable as well as the independent variables using the instruments. We then proceed to illustrate the finite sample properties of all the estimators we consider using the DTUS data, which tracks the respondents' activities for an entire week, and hence is a valuable benchmark. Multiple empirical findings are also drawn from the DTUS data. Finally, we compare the estimated labour supply elasticities using the ATUS impute estimator and that using the CPS recalled hours, and we are able to get a number of interesting empirical findings that are new in the labour economics literature.

Table 1: Simulations Based on the Dutch Time Use Survey (DTUS)

$\mathrm{Corr}(\tilde{X}_i, U_i)$ / $\mathrm{Corr}(\tilde{X}_i, \tilde{Z}_i)$		Panel A: $n = 250$					Panel B: $n = 500$				
		$\hat{\beta}_{re}$	$\hat{\beta}_{wk}$	$\hat{\beta}_{im}$	$\hat{\beta}_{pool}$	$\hat{\beta}_{day}$	$\hat{\beta}_{re}$	$\hat{\beta}_{wk}$	$\hat{\beta}_{im}$	$\hat{\beta}_{pool}$	$\hat{\beta}_{day}$
0 / 1	MSE	1.255	0.004	0.051	0.122	0.051	1.241	0.002	0.023	0.061	0.023
	Bias^2	1.232	0.000	0.000	0.000	0.000	1.230	0.000	0.000	0.000	0.000
	Var	0.023	0.004	0.051	0.122	0.051	0.011	0.002	0.023	0.061	0.023
0.25 / 0.95	MSE	1.245	0.002	0.049	0.125	0.049	1.240	0.001	0.022	0.061	0.022
	Bias^2	1.222	0.000	0.000	0.000	0.000	1.228	0.000	0.000	0.000	0.000
	Var	0.024	0.002	0.049	0.125	0.049	0.012	0.001	0.022	0.061	0.022
0.5 / 0.80	MSE	1.248	0.005	0.052	0.126	0.070	1.245	0.003	0.023	0.059	0.028
	Bias^2	1.222	0.000	0.000	0.000	0.001	1.231	0.000	0.000	0.000	0.000
	Var	0.027	0.005	0.052	0.126	0.069	0.013	0.003	0.023	0.059	0.028
0.75/ 0.43	MSE	1.234	0.079	0.127	0.207	639.291	1.231	0.036	0.058	0.098	12.010
	Bias^2	1.182	0.001	0.001	0.002	0.326	1.206	0.000	0.000	0.000	0.091
	Var	0.052	0.077	0.126	0.205	638.965	0.025	0.036	0.058	0.098	11.919
$\mathrm{Corr}(\tilde{X}_i, U_i)$ / $\mathrm{Corr}(\tilde{X}_i, \tilde{Z}_i)$		Panel C: $n = 1000$					Panel D: $n = 2500$				
		$\hat{\beta}_{re}$	$\hat{\beta}_{wk}$	$\hat{\beta}_{im}$	$\hat{\beta}_{pool}$	$\hat{\beta}_{day}$	$\hat{\beta}_{re}$	$\hat{\beta}_{wk}$	$\hat{\beta}_{im}$	$\hat{\beta}_{pool}$	$\hat{\beta}_{day}$
0 / 1	MSE	1.235	0.001	0.011	0.030	0.011	1.230	0.000	0.004	0.012	0.004
	Bias^2	1.229	0.000	0.000	0.000	0.000	1.228	0.000	0.000	0.000	0.000
	Var	0.006	0.001	0.011	0.030	0.011	0.002	0.000	0.004	0.012	0.004
0.25 / 0.95	MSE	1.128	0.000	0.011	0.029	0.010	1.231	0.000	0.004	0.012	0.004
	Bias^2	1.232	0.000	0.000	0.000	0.000	1.229	0.000	0.000	0.000	0.000
	Var	0.006	0.000	0.011	0.029	0.010	0.002	0.000	0.004	0.012	0.004
0.5 / 0.80	MSE	1.233	0.001	0.011	0.030	0.013	1.230	0.001	0.004	0.012	0.005
	Bias^2	1.226	0.000	0.000	0.000	0.000	1.228	0.000	0.000	0.000	0.000
	Var	0.007	0.001	0.011	0.030	0.013	0.003	0.001	0.004	0.012	0.005
0.75/ 0.43	MSE	1.229	0.017	0.027	0.047	0.087	1.229	0.007	0.011	0.018	0.021
	Bias^2	1.217	0.000	0.000	0.000	0.011	1.224	0.000	0.000	0.000	0.001
	Var	0.012	0.017	0.027	0.047	0.076	0.005	0.007	0.011	0.018	0.020

[1] This table compares finite sample performance of various estimators using the DTUS data. $10,000$ random samples of different sizes are drawn from the original DTUS sample of $6,567$ individual-year records.

[2] The two numbers in the first column represent: (i) correlation coefficient between regressor $\tilde{X}_i$ and error term U_i (degree of endogeneity); (ii) correlation coefficient between regressor $\tilde{X}_i$ and IV $\tilde{Z}_i$ (strength of IV). Both are adjusted by changing the parameter ρ in the simulation setup.

[3] $\hat{\beta}_{re}$ is the 2SLS estimator using the error-ridden recalled weekly hours worked in the DTUS. $\hat{\beta}_{re}$ exhibits large bias.

[4] $\hat{\beta}_{wk}$ is the 2SLS estimator given in equation (3.5), which uses the accurate weekly hours worked in the DTUS and serves as an infeasible benchmark for the three estimators based on the ATUS. $\hat{\beta}_{wk}$ has virtually no bias and the smallest variance.

[5] For each individual in the DTUS, we randomly draw one from the seven days using the diary day sampling probabilities of the ATUS, thus obtained samples that imitate the ATUS, and we apply $\hat{\beta}_{im}$, $\hat{\beta}_{pool}$ and $\hat{\beta}_{day}$ to them in order to evaluate their performance.

[6] $\hat{\beta}_{im}$ has virtually no bias and the smallest variance among the three, followed by $\hat{\beta}_{pool}$.

[7] $\hat{\beta}_{day}$ is numerically equivalent to $\hat{\beta}_{im}$ when $\tilde{X}_i$ is exogenous. When $\tilde{X}_i$ is endogenous, however, $\hat{\beta}_{day}$ could display notable bias and enormous variance, especially when the sample size is smaller (and hence each day subsample is even smaller).

Table 2: Weekly Labour Supply Elasticity Estimates (Hundredths): the DTUS

	Married Men			Married Women		
	$\hat{\beta}_{re}$	$\hat{\beta}_{wk}$	$\hat{\beta}_{im}$	$\hat{\beta}_{re}$	$\hat{\beta}_{wk}$	$\hat{\beta}_{im}$
n of kids aged < 18	0.93	0.39	0.22	0.02	-16.81	-21.02
	(0.41)	(0.58)	(1.19)	(0.80)	(1.72)	(3.27)
			[1.16]			[3.20]
Educ: completed 2ndry	2.14	-1.16	-7.43	-2.12	11.88	9.79
	(1.12)	(1.59)	(3.11)	(2.09)	(4.47)	(8.27)
			[3.14]			[8.17]
Educ: above 2ndry	4.13	-2.06	-5.59	-0.86	22.68	21.53
	(1.19)	(1.68)	(3.04)	(2.48)	(5.32)	(9.68)
			[3.32]			[9.81]
P value of joint Hausman test	0.00	0.14		0.00	0.50	
n of Obs.	1746	1746	1746	835	835	835
R squared[4]	0.06	0.03	0.07	0.18	0.39	0.26

[1] The other control variables are age, age-squared, a dummy of working in private sector, an urban area dummy, and year dummies.

[2] $\hat{\beta}_{re}$ uses the recalled weekly hours; $\hat{\beta}_{wk}$ uses the true diary weekly hours; $\hat{\beta}_{im}$ uses the sample where only one day is randomly chosen for each individual using the ATUS diary day sampling weights.

[3] Standard errors are in parentheses. For $\hat{\beta}_{im}$, we report the standard errors using the formulas in Theorem 4 (under Assumption 5, in round parentheses) and the formulas in Supplementary Appendix A (in square parentheses) using the DTUS correlation matrix.

[4] We conduct the joint Hausman tests (i.e. the coefficients associated with the three regressors in the table) regarding whether there are significant differences between $\hat{\beta}_{re}$ and $\hat{\beta}_{im}$, and between $\hat{\beta}_{wk}$ and $\hat{\beta}_{im}$, respectively.

[5] The R squared for impute estimator is the average R squared of the seven linear regression of daily hours worked $H_{it} = X_i'\beta_t + U_{it}$ for $t = 1, \ldots, 7$.

Table 3: Weekly Labour Supply Elasticity Estimates: the CPS and the ATUS

Panel A: Mean and std dev of hours and wage				
	Married Men	Unmarried Men	Married Women	Unmarried Women
CPS Usual Weekly Hours Worked[1]	39.63	38.42	32.50	35.52
s.d.	(6.13)	(7.26)	(10.43)	(8.63)
ATUS Hours Worked on Diary Day	4.70	4.74	3.56	4.18
s.d.	(4.55)	(4.44)	(4.00)	(4.21)
ATUS Imputed Weekly Hours Worked	41.27	40.38	31.96	36.18
s.d. (lower bound)[2]	(9.57)	(9.79)	(9.26)	(9.68)
Hourly Wage (2017 US dollars)	21.88	18.65	18.70	16.56
Panel B: Elasticities (hundredths)[3]				
	Married Men	Unmarried Men	Married Women	Unmarried Women
Wage (CPS)	5.39	11.38	15.89	11.72
	(0.89)	(1.06)	(1.26)	(1.07)
Wage (ATUS)	1.47	4.71	10.48	8.14
	(3.27)	(3.26)	(3.25)	(3.24)
	[3.31]	[3.22]	[3.20]	[3.13]
Spouse weekly earnings (CPS)	−0.19		−9.43	
	(0.41)		(0.77)	
Spouse weekly earnings (ATUS)	−3.47		−5.79	
	(1.60)		(2.08)	
	[1.57]		[2.03]	
Num. of kids age < 5 (CPS)	−0.80		−8.58	
	(0.48)		(0.82)	
Num. of kids age < 5 (ATUS)	−1.08		−8.97	
	(1.91)		(2.08)	
	[1.92]		[2.12]	
Num. of kids ages 5–18 (CPS)	−0.00		−2.87	
	(0.26)		(0.42)	
Num. of kids ages 5–18 (ATUS)	−0.44		−1.20	
	(1.10)		(1.16)	
	[1.09]		[1.17]	
R squared (CPS)	0.08	0.15	0.22	0.15
R squared (ATUS)[6]	0.16	0.24	0.17	0.17
p value of joint Hausman test	0.24	0.05	0.06	0.27
n of obs.	3889	3816	5602	5731

[1] This is the number of hours per week that the respondent usually works at his/her current job at the reported hourly wage rate.
[2] See footnote 40 in the paper for more details.
[3] The estimates based on the CPS recalled weekly hours are $\hat{\beta}_{re}$; the estimates based on the ATUS diary day hours are $\hat{\beta}_{im}$.
[4] The standard errors of estimating elasticities using CPS recalled hours are in parentheses. When use ATUS hours, we report the standard errors using the formulas in Theorem 4 (under Assumption 5, in round parentheses) and the formulas in Supplementary Appendix A (in square parentheses) using the DTUS correlation matrix.
[5] The elasticities are evaluated at the respective mean hours worked in each data source.
[6] The R squared for impute estimator is the average R squared of the seven linear regression of daily hours worked $H_{it} = X_i'\beta_t + U_{it}$ for $t = 1, \ldots, 7$.
[7] For each sample group, we conduct joint Hausman tests regarding whether there are significant differences between $\hat{\beta}_{re}$ and $\hat{\beta}_{im}$.
[8] The other control variables are age, age-squared, the number of children aged below 5, the number of children aged between 5 and 18, two education dummies, eight Census division dummies, a metropolitan area dummy, race dummies, year dummies, occupation dummies and industry dummies.

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